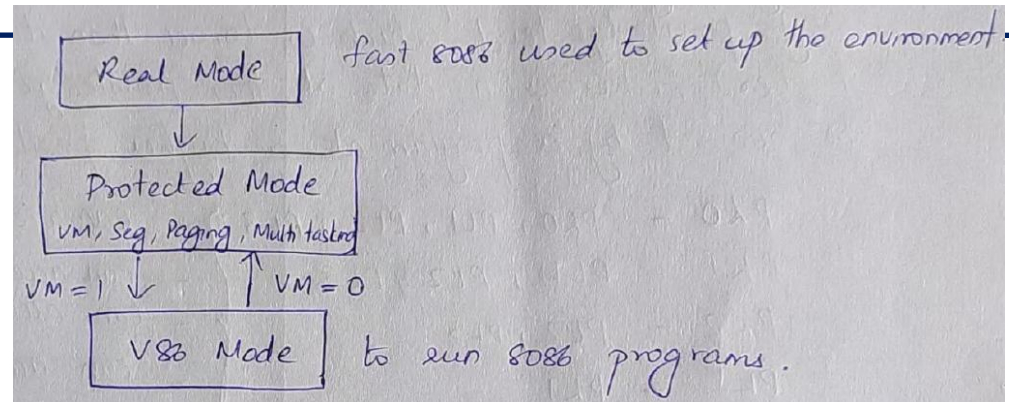


Operating Modes of 80386

- 80386 can operate in 3 modes
- 1. Real Mode
- 2. Protected mode
- 3. Virtual Mode



- 1. Real Mode: It is the default mode selected when 80386 is reset. In real mode, 80386 does not behave like a 80386 but it behaves like a 8086, it just works like a fast 8086 due to 20 MHz clock of 80386. All the registers are just like 8086, even the memory used is only 1MB, just like 8086. PA = Segment address X 10H + Offset address. This mode is basically used to run BIOS on reset.
- 80386 starts in real mode and it creates the environment for protected mode i.e. it initializes all the registers and tables like GDT, LDT page table etc. so that the protected mode can work
- Once the required registers are initialized, we can switch to protected mode by making the protection Enable (PE bit) = 1 in CR0 (Control register 0)
- We can't switch back to real mode from protected mode. But we can go to Virtual 8086 mode by making VM bit in flag register 1, here 8086 programs can run and by making VM=0, we can go back to protected mode

Real Mode of 80386

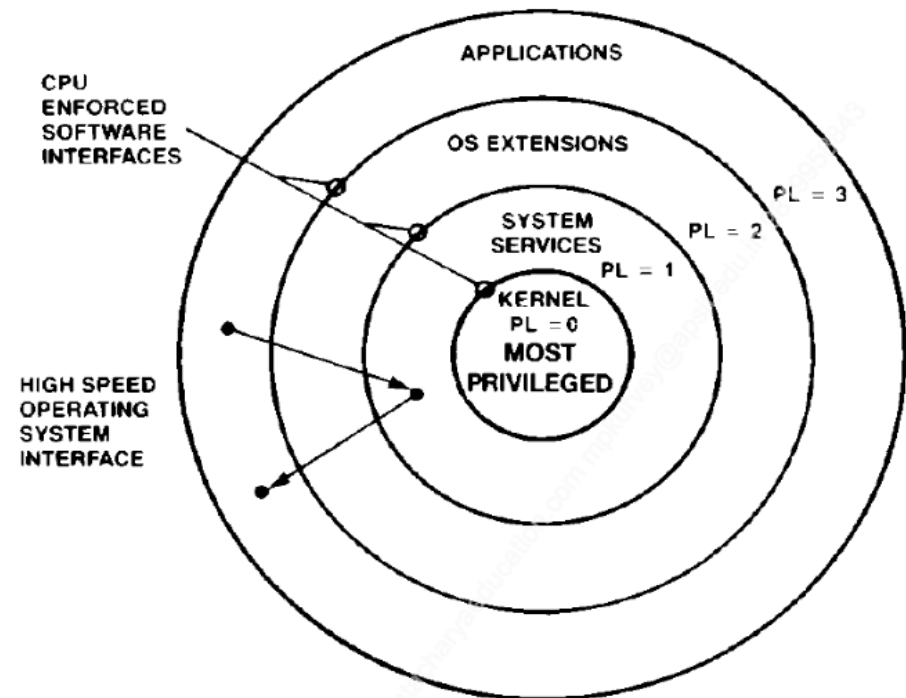
- It is the **default mode selected when 80386 is reset**. In this mode, 80386 μP simply behaves as a fast 8086 machine. All registers are just like 8086. Even the memory used is only 1 MB, just like in 8086. Physical address calculation is also like in 8086. (Physical Address = Base address X 10H + Offset Address. **This mode is basically used to run the monitor program(BIOS) on reset.** Once the required registers are initialised, we can switch to protected mode by making PE bit =1 in CR0.
- **Segment Registers:** There are 6, 16-bit segment registers containing the base addresses of their respective segments. The registers are CS,SS,DS,ES,FS and GS
- **Offset Registers:** There are 5, 16-bit offset registers containing the offset addresses for various segments. The registers are IP,SP,BP,SI and DI. The 32-bit extended form(ESI, EDI etc) of these registers is not used in Real Mode
- **Data Registers:** There are 4, 16-bit data registers used as General purpose Registers during programming. The registers are AX,BX,CX and DX. They can be also used as 8, 8-bit registers AL,AH,BL,BH,CL,CH,DL and DH. The 32-bit extended form(EAX,EBX etc) of these registers is not used in Real Mode

Real Mode of 80386

- **Flags:** Only the lower 12 -bits of the Flag Register are used in the Real Mode. The 32-bit extended form of Flag Register i.e. EFLAGS is available only in protected Mode.
 - **Control Registers:** Only the LSB of CR0 is available in Real Mode. The remaining bits of CR0 and the remaining control registers are available only in protected Mode. The LSB of CR0 is “PE” bit which should be made = “1” to begin protected Mode.
 - **Debug Registers and Test Registers:** These registers are not available in Real Mode
 - **Memory Range:** The size of memory available in Real Mode is 1 MB and has address range from 00000HFFFFFFH, just like in 8086.
 - **I/O Range:** A total of 64 K I/O addresses are available having a range from 0000H.....FFFFH just like in 8086
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Protected Mode of 80386

- 80386 μP provides a very advanced mode of operations called the protected Mode. In protected Mode, 80386 μP provides dedicated hardware to prevent user programs from affecting other user programs and also safeguards the operating system from being affected by user programs.
- There are four privilege levels, assigned to programs and data to define their privileges.



Protected Mode of 80386

- **Level 0:** This level is assigned to the operating system Kernal(Main part of the operating system). It is the most privileged level. Any program at this level can access all the data at any privilege level, whereas a data at this privilege level can only be accessed by a program at privilege level 0.
 - **Level 1:** This level is assigned to the system Services such as File Handling, Device Drivers. It is the second most privilege level. Any program at this level can access the data at any privilege level which is lower than this level(numerically higher), whereas a data at this privilege level can only be accessed by a program at privilege level 0 or privilege level 1.
 - **Level 2:** This level is assigned to the Custom Extensions of the OS. It is the third most privilege level. Any program at this level can access the data at any privilege level which is lower than this level(numerically higher), whereas a data at this privilege level can only be accessed by a program at privilege level 0 or privilege level 0,1 or 2.
 - **Level 3:** This level is assigned to all the user Application and Programs. It is the least privilege level. Any program at this level can normally access the data at privilege level 0, whereas a data at this privilege level can only be accessed by a program at privilege level 0 or privilege level 0...3.
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Comparison of Real and Protected Mode

Real Mode	Protected Mode
16- bit operations, Default mode after reset	32- bit operations, Protected mode entered after making LSB of CR0 (PE=1)
Four 16-bit General Purpose registers (AX, BX, CX, DX)	Four 32-bit General Purpose registers (EAX, EBX, ECX, EDX)
Flag Register: 16-bit	Flag Register: 32-bit (EFLAGS)
Control register: Only LSB of CR0 is available in real mode	All four control registers are available CR0, CR1, CR2, CR3
PA = Seg Add X 10H + Offset add	Complex segment and page translation
16 bit offset, therefore maximum size of Segment is 64 KB	32 bit offset, therefore maximum size of Segment is 4 GB
Memory range: Size of memory available in real mode is 1MB and has address range 00000HFFFFFH, just like in 8086	Total physical memory $2^{32} = 4 \text{ GB}$ Virtual Memory = $2^{46} = 64 \text{ TB}$